NNN-P-395D
February 24, 1978
SUPERSEDING
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March 13, 1970

FEDERAL SPECIFICATION

PIPET, VOLUMETRIC (TRANSFER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass volumetric pipets (transfer) (see 6.1).

1.2 Classification.

1.2.1 Classes and capacities. Pipets covered by this specification shall be of the following classes and capacities as specified (see 6.2):

1.2.1.1 Classes.

Class A - Serialized - precision.
Class A - Nonserialized - precision.
Class B - General purpose.

1.2.1.2 Capacities: 0.5, 1, 2, 3, 4, 5, 10, 20, 25, 50, 100, 200, milliliters (ml).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

DD-G-541 - Glass (Laboratory).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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FSC 6640

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

C 148 - Polariscopic Examination of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards Internal Report:

NBSIR 74-461 - The Calibration of Small Volumetric Laboratory Glassware.

(Application for copies should be addressed to the National Technical Information Service (NTIS), Springfield, VA 22151.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Borosilicate glass. Pipets shall be transparent borosilicate glass.

3.2 Annealing. Ring strain shall be no greater than that of temper no. 1 when tested as specified in 4.3.1 except as permitted for the tempered tip in 3.6.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.3 Capacity tolerance.

3.3.1 Class A serialized and nonserialized. Class A serialized and nonserialized pipets shall be calibrated to deliver (symbol: TD) at 20°C and shall have capacity tolerances in accordance with the requirements of table I when tested as specified in 4.3.2.

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TABLE I. Capacity tolerances and dimensions for class A, serialized and nonserialized

Capacity (ml)	Capacity tolerance (± ml)	Length of suction tube (Min., mm)	Length of delivery tube (Min., mm)	Delivery and suction tube wall thickness (Min., mm)	Inside diameter at capacity mark (Max., mm)
1/2	0.005	150	110	1.2	4
1	0.007	150	110	1.2	4
2	0.01	150	125	1.2	4
3	0.01	150	140	1.2	4
4	0.01	150	145	1.2	4
5	0.015	150	145	1.2	4
10	0.02	160	160	1.2	4
20	0.03	170	210	1.2	5
25	0.03	170	220	1.2	5
50	0.05	170	230	1.2	6
100	0.08	170	240	1.2	7
200	0.1	170	240	1.2	7

3.3.2 Class B. Class B pipets shall be calibrated to deliver (symbol: TD) at 20° C and shall have capacity tolerances in accordance with requirements of table II when tested as specified in 4.3.2.

TABLE II. Capacity tolerances and dimensions for class B

Capacity (ml	Capacity tolerance (± ml)	Length of suction tube (Min., mm)	Length of delivery tube (Min., mm)	Delivery and suction tube wall thickness (Min., mm)	Inside diameter at capacity mark (Max., mm)
1/2	0.01	150	110	1.2	4
1	0.015	150	110	1.2	4
2	0.02	150	125	1.2	4
3	0.02	150	140	1.2	4
4	0.02	150	145	1.2	4
5	0.03	150	145	1.2	4
10	0.04	160	160	1.2	4
20	0.06	170	210	1.2	5.5
25	0.06	170	220	1.2	5.5
50	0.1	170	230	1.2	6
100	0.16	170	240	1.2	7
200	0.2	170	240	1.2	7

3.4 Dimensions.

3.4.1 Class A serialized and nonserialized. Class A serialized and nonserialized pipets shall have the dimensions as specified in table I.

3.4.2 Class B. Class B pipets shall have the dimensions as specified in table II.

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3.5 Markings.

3.5.1 Capacity line. The pipet shall have a capacity line positioned so as to deliver, as specified, the quantity indicated by the numerical marking and marked method of use. The capacity line shall be a sharply defined line of uniform width. The line width shall not exceed 0.4 mm or vary more than 0.1 mm. Within at least two opposite quarters, the bottom edge of the line on the front and the top edge of the line on the back shall be straight. Capacity lines shall be perpendicular to the vertical axis of the pipet and not more than 6 cm from top of bulb. The capacity line shall be applied by one of the following methods: etched and filled with a permanent pigment; etched through a vertical color band fired into the glass or by application of an enamel fired onto the color band without etching; or by application of a stain which is fired into the glass without etching; or by application of an enamel fired into the glass without etching. The capacity line shall extend at least 3/4 way around the suction tube. The break in the line should be located approximately 180° from the capacity marking on the pipet.

3.5.2 Identification marking. All pipets shall be permanently and legibly marked with the manufacturer name or trademark, the capacity, the temperature at which the graduations apply (20° C) and the method of use (to deliver). Each class A serialized pipet shall be marked with a serial number. Class A pipets shall be marked with the letter A to signify compliance with applicable construction and accuracy requirements. There shall be no change in appearance of capacity line or marking (see 3.5.1 and 3.5.3) when tested as specified in 4.3.4. Pipets calibrated to deliver (TD) shall be identified by a wide band or two narrow bands permanently marked on the mouth piece of the pipet as indicated in 3.8.

3.6 Delivery tip. Delivery tip shall be made with a gradual taper of 1.5 to 3 centimeters (cm). The end of the tip shall be perpendicular to the longitudinal axis of the tip. The outside edge of the tip shall be beveled slightly at the end and the bevel shall be ground or fire polished. A nonuniform constriction at the bore of the orifice shall not be acceptable. The tolerances of delivery time shall be within the limits of table III for class B and table IV for both class A serialized and nonserialized and shall be not longer than 1 minute. Limits of error in marked delivery time for class A serialized shall be as specified in table V.

3.6.1 Tempered tip. The pipet shall have a tempered tip. When tested in index oil which matches the refractive index of the glass being used and in accordance with 4.3.1, the temper shall be between 75 and 220 millimicrons.

TABLE III. Tolerance for delivery time class B

Size (ml)	Delivery time sec	
	Min.	Max.
1/2	3	20
1	3	20
2	3	20
3	5	20
4	5	20
5	8	25
10	8	25
20	10	30
25	15	35
50	15	35
100	20	40
200	35	60

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TABLE IV. Minimum delivery time for class A serialized and nonserialized pipets (maximum delivery time 60 sec)

Capacity (in ml) up to and including	Delivery time (in seconds)
0.5	10
1	10
2	10
3	10
4	15
5	15
10	15
20	25
25	25
50	30
100	40
200	50

TABLE V. Limits of error in marked delivery time class A serialized

Delivery time (in seconds) up to and including	Limit of error in marked delivery time
	sec
15	3
20	4
30	6
50	8
100	15
200	20

3.7 Workmanship. The pipet shall be transparent and free from striae, surface irregularities and other defects which may distort the appearance of the liquid surface or the portion of the graduation line seen through the glass. The pipet shall also be free of blisters, boils, seeds, stones, and other defects which may weaken the glass, or impair the usage. Top of suction tube shall be finished with a smooth plane fire polished top, at right angles to axis.

3.8 Color-coding. The pipet shall be marked with one or two color bands as specified (see table VI). The color of the band shall be bright and distinct. When tested as specified in 4.3.4, the color shall show no damage other than loss of gloss. Banding shall meet the alkali and acid test as specified in 4.3.4.1 and 4.3.4.2 respectively.

TABLE VI. Color-coding

Size ml	Color coding band
1/2	2 black
1	blue
2	orange
5	white
10	red
20	yellow
25	blue
50	red
100	yellow
200	blue

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling for visual examination. Sampling for visual examination shall be in accordance with MIL-STD-105, inspection level II, and the classification of defects of AQL's indicated in table VII.

4.2.1 Sampling for tests.

4.2.1.1 Capacity accuracy and delivery time. Sampling shall be in accordance with MIL-STD-105, inspection level S-4. For class A serialized and nonserialized, an AQL of 1.0 percent defective shall apply to the two tests combined. For class B, an AQL of 2.5 percent defective shall apply to the two tests combined (see 4.3.2 and 4.3.3).

4.2.1.2 Annealing and permanency of marking tests. A sample in accordance with MIL-STD-105, inspection level S-4 shall be taken from each lot. Each pipet in the sample shall be tested for annealing (see 4.3.1), prior to permanency of marking tests. A different sample, in accordance with MIL-STD-105, inspection level S-1, shall be used for the tests as specified in 4.3.4.1 and 4.3.4.2. An AQL of 0.65 percent defective shall apply to the two tests combined.

TABLE VII. Classification of defects for visual examination

Categories	Defects	Ref. para.
Class A, serialized		
Major:	AQL 1.5 percent defective	
	"A" mark or serial number missing	3.5.2
	Marking missing or improper	3.5.2 3.8
	Chips and cracks	3.7
	Improper delivery tip	3.6
	Smears of filler, scratches, acid and other blemishes in reading area which interfere with setting	3.5.1 3.5.2 3.7
	Capacity or graduation lines of wrong width or not straight (where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain)	3.5.1
Class A, nonserialized		
Major:	AQL 4.0 percent defective	
	"A" mark missing	3.5.2

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TABLE VII. Classification of defects for visual examination (con.)

Categories	Defects	Ref. para.
Class A, nonserialized and class B		
Major: AQL 4.0 percent defective		
	Mark missing or improper	3.5.2 3.8
	Chips and cracks	3.7
	Smears of filler, scratches, and burns, and other blemishes in reading area which interfere with setting	3.5.1 3.5.2 3.7
	Capacity or graduation lines of wrong width or not straight (where lines are filled with stain and the stain extends beyond the edge of the etched line, measure the width of the stain)	3.5.1
	Improper delivery tip	3.6

4.2.2 Examination of preparation for delivery requirements. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table VIII. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for examination at one time. Sampling shall be in accordance with MIL-STD-105, and the inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 expressed in terms of percent defective.

TABLE VIII. Examination of preparation for delivery

Examine	Defects
Container	Not as specified.
Material	Any component missing, damaged or not as specified.
Content	Not as specified.
Workmanship	Bulging or distortion of containers. Cushioning inadequate or improper.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.

4.3 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.3.1 Annealing. Pipets shall be tested in accordance with method A or B of ASTM C 148.

4.3.2 Capacity accuracy. Class A serialized and class A nonserialized and class B pipets shall be tested for capacity accuracy in accordance with the procedures of NBSIR 74-461.

4.3.3 Delivery time. Delivery time of class A serialized, class A nonserialized and class B pipets shall be as specified in table III, table IV, and table V.

4.3.4 Permanency of marking. A different pipet shall be used for each of the following tests.

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4.3.4.1 Resistance to alkali. The pipet shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The pipet shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth (see 3.8).

4.3.4.2 Resistance to acid. The pipet shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The pipet shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth (see 3.8).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit container. One pipet shall be packaged in a fiberboard box or in a tube commonly used for the item. The box shall be provided with fiberboard blocking or sufficient cushioning material to prevent shifting of contents. The box shall be closed with adhesive, staples or tape to secure the flaps.

5.1.1.2 Intermediate container. Eighteen pipets of the same type, class and size packaged as specified in 5.1.1.1, shall be intermediately packaged in a corrugated fiberboard box. The box shall be closed with adhesive, staples or tape to secure the flaps.

5.1.2 Commercial. The pipets shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment handling and storage.

5.2 Packing. Packing shall be Level A, or Commercial as specified (see 6.2).

5.2.1 Level A. Sixteen intermediate packages of pipets of one class and capacity of 10 ml or smaller or eight intermediate packages of one class and capacity of pipets larger than 10 ml shall be packed in a weather resistant, corrugated fiberboards box. The box shall be closed with adhesive, staples or tape to secure the flaps. The box shall be sealed to enhance water resistance. The box shall be banded with metal or plastic strapping or tape for reimforcement. Closure of the box shall be as specified in the appendix of PPP-B-636.

5.2.2 Commercial. The packaged pipets shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Marking. The unit, intermediate and shipping containers and unitized loads, as applicable shall be marked as specified in the contract or order (see 6.2).

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6. NOTES

6.1 Intended use. Pipets covered by this specification are intended for laboratory applications. In general, class B pipets will be most applicable for Federal and Military agencies. Class A serialized pipets are intended for precision measurements and are of a quality such that specific pipets will be acceptable by the National Bureau of Standards for calibration at specified points if such calibration is justified by the nature of the usage.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Packaging and packing required (see 5.1 and 5.2).
(c) Class and capacity required (see 1.2.1).
(d) Marking required (see 5.4).

MILITARY CUSTODIANS:

Army - MD
Navy - MS
Air Force - 03
DOD - MD
DLA - DM

Review activities:

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Air Force - 03
DOD - MD
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Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

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